

BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

$Z(\rightarrow \ell\ell) + \text{b/c boosted jets in Run 3}$

First look at Summer22 $DY \rightarrow \ell\ell$ MC: samples, cutflow and kinematics

7 October 2026

Campaign and setup

Goal: measure $\sigma(Z+b) / \sigma(Z+c)$ with a boosted AK8 jet recoiling against $Z \rightarrow ee/\mu\mu$. First step: tune the selection on $DY \rightarrow \ell\ell$ MC.

MC CAMPAIGN

Summer22

Run3Summer22 = 2022 eras
C-D (pre-EE)

FORMAT

NanoAODv12

130X MC conditions

ENERGY ($\sqrt{s}$)

13.6 TeV

Run 3 pp collisions

NORMALIZATION

7.98 fb⁻¹

2022 C-D golden JSON;
brilcalc check pending

MC only, no data yet. Processed: 1 file per sample (test run). Code: BTVNanoCommissioning, branch coffea_machine, commit f8cd578.

Summer22 $DY \rightarrow \ell\ell$ MC · 1 file per sample · 2

MC samples: $DY \rightarrow \ell\ell$, stitched in $pT(\ell\ell)$

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/DYto2L-2Jets_MLL-50[_PTLL-X]_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/  
Run3Summer22NanoA0Dv12-130X_mcRun3_2022_realistic_v5-vN/NANOA0DSIM
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Sample	LHE $pT(\ell\ell)$ used	σ [pb]	DAS files	DAS events	Test: events used
Inclusive (MLL-50)	< 100 GeV	6346	525	93.8 M	197,672
PTLL-100	100–200 GeV	105.0	418	56.4 M	161,921
PTLL-200	200–400 GeV	11.24	335	44.7 M	91,186
PTLL-400	400–600 GeV	0.643	424	41.6 M	2,727
PTLL-600	> 600 GeV	0.0994	185	9.50 M	939

PTLL-X samples are inclusive above X, so each is kept to its own LHE $pT(\ell\ell)$ slice. $\sigma(\text{PTLL-X}) = 6346 \text{ pb} \times \text{fraction of inclusive events with } pT(\ell\ell) > X$ (measured on 27.7 M inclusive events). "Test: events used" = events of 1 file inside the slice.

Event selection, per channel

Z→ee / Z→μμ

- Trigger: HLT_Ele30_WPTight_Gsf / HLT_IsoMu24
- Leptons $p_T > 25$ GeV, $|\eta| < 2.4$; leading $p_T > 35$ GeV
- e: tight cut-based ID, IP cuts, EB–EE gap veto
- μ : tight ID, PF isolation < 0.15
- Opposite sign (OS)
- $71 < m_{\ell\ell} < 111$ GeV
- $\text{PuppiMET} < 50$ GeV

Leading AK8 jet ("jet" step)

- $\Delta R > 0.8$ from selected leptons
- Tight jet ID, two soft-drop subjets
- $p_T > 200$ GeV, $|\eta| < 2.5$, $m_{SD} > 40$ GeV

Weights and flavour

- $\text{genWeight} \times \text{pileup} \times \text{lepton ID/iso/reco SFs}$
- No trigger SF yet
- Jet flavour from B/C hadron counts: bb, b, cc, c, light

Cutflow, $Z \rightarrow ee$: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	197,672	32,660	9,802	9,745	9,396	8,872	1	0.01%
PTLL-100	161,921	46,448	13,156	13,037	12,032	10,642	139	1.3%
PTLL-200	91,186	31,202	11,594	11,431	10,398	8,584	1,211	14.1%
PTLL-400	2,727	1,013	458	447	396	287	77	26.8%
PTLL-600	939	229	101	100	82	54	12	22.2%

1,440 raw MC events pass the full $Z \rightarrow ee$ selection; 84% of them come from PTLL-200. The boosted-jet step keeps 1.3% (PTLL-100) to 14% (PTLL-200) of Z events.

Cutflow, $Z \rightarrow \mu\mu$: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	197,672	42,062	17,685	17,685	17,054	16,139	8	0.05%
PTLL-100	161,921	48,592	20,606	20,604	19,043	16,788	225	1.3%
PTLL-200	91,186	31,717	16,868	16,865	15,258	12,453	1,753	14.1%
PTLL-400	2,727	1,048	659	659	592	407	104	25.6%
PTLL-600	939	319	199	199	178	102	31	30.4%

2,121 raw MC events pass the full $Z \rightarrow \mu\mu$ selection; 83% of them come from PTLL-200. Jet-step efficiencies similar to $Z \rightarrow ee$.

Yields by AK8 jet flavour (7.98 fb⁻¹)

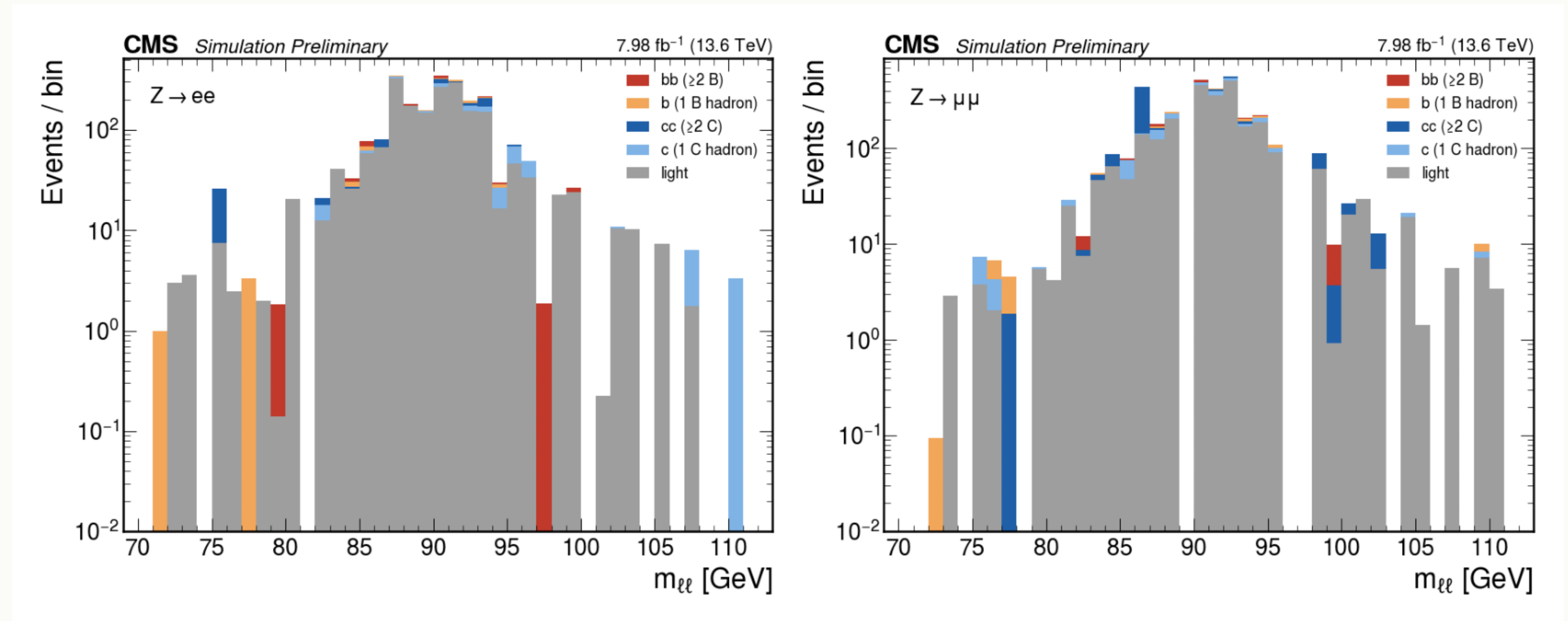
Channel	light	c	cc	b	bb	total
Z→ee	1897.8	137.0	128.7	65.1	31.1	2259.7
fraction	0.840	0.061	0.057	0.029	0.014	
Z→μμ	2228.6	265.0	415.5	−191.8	67.3	2784.7
fraction	0.800	0.095	0.149	−0.069	0.024	

Heavy-flavour numbers are not reliable yet

With 1 file per sample each inclusive-sample event weighs about 250, and amcatnlo events can have negative weights: one or two such events give the negative b yield in Z→μμ. The b and c fractions need the full samples.

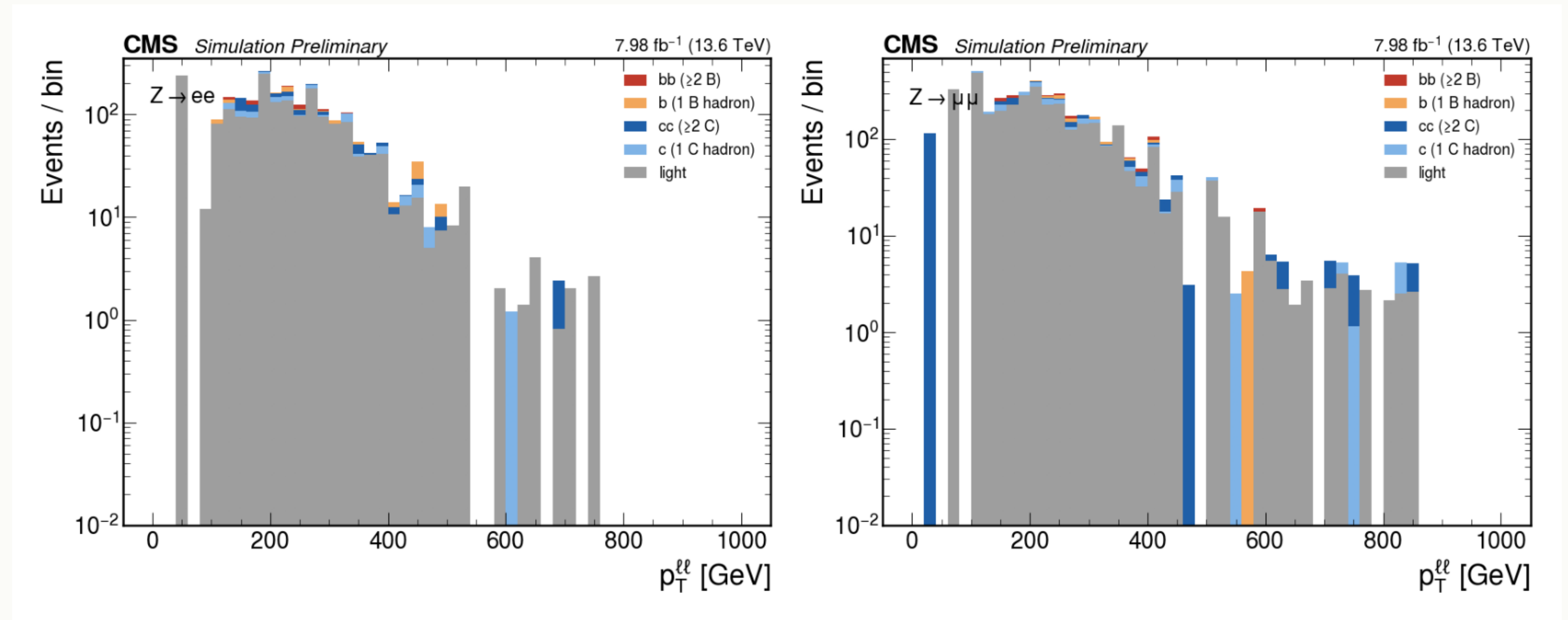
Dilepton mass

$m(\ell\ell)$ after the full selection (71-111 GeV window): peaks at the Z mass in both channels.



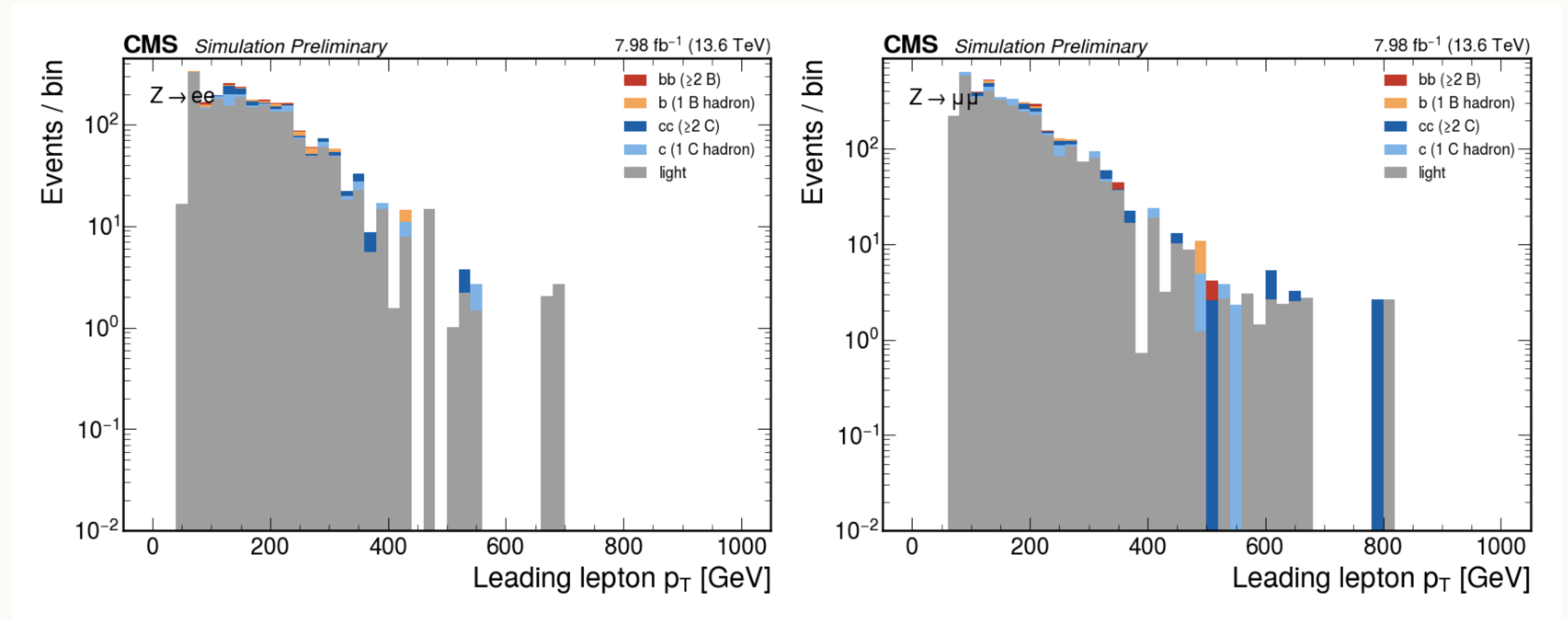
Dilepton p_T

$p_T(\ell\ell)$ of the Z candidate after the full selection; the Z recoils against the boosted AK8 jet.



Leading-lepton p_T

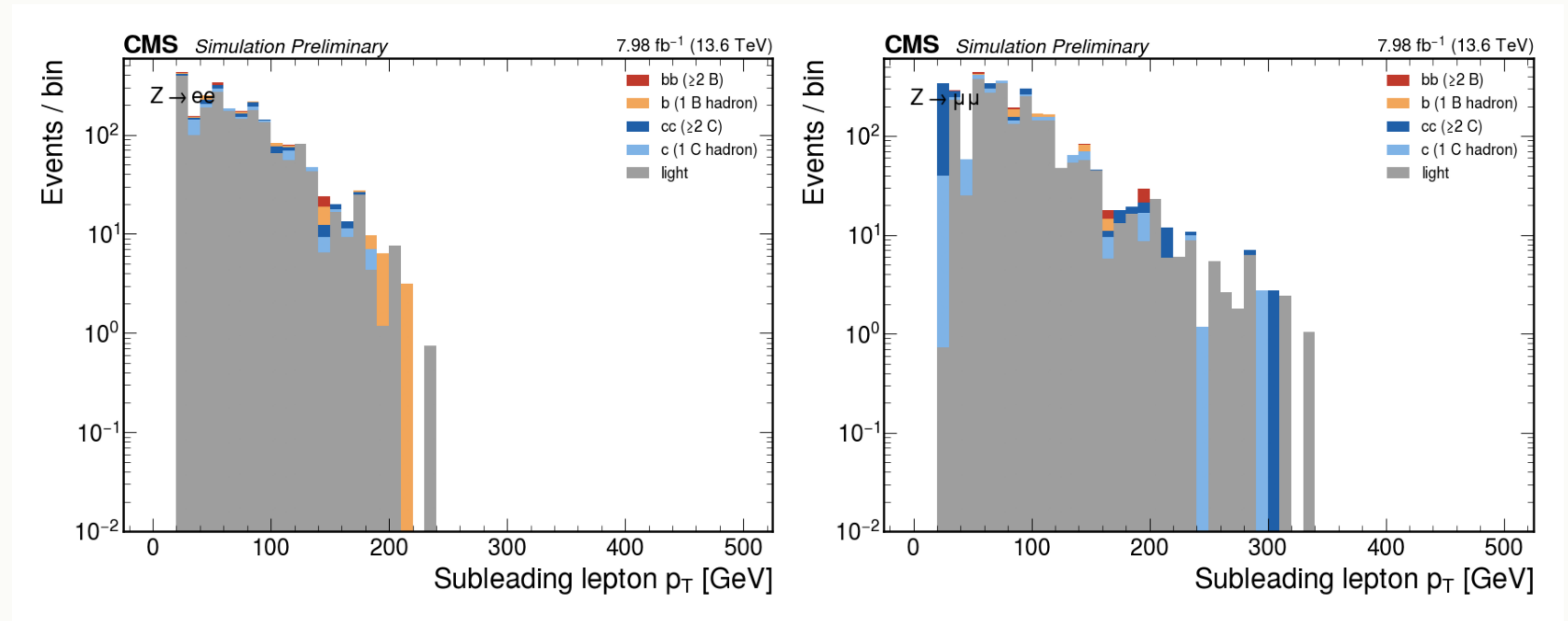
Leading lepton of the Z candidate, after the full selection ($p_T > 35$ GeV required).



Summer22 DY $\rightarrow\ell\ell$ MC · 1 file per sample · stacked by AK8 jet flavour · Z $\rightarrow ee$ left, Z $\rightarrow\mu\mu$ right · 10

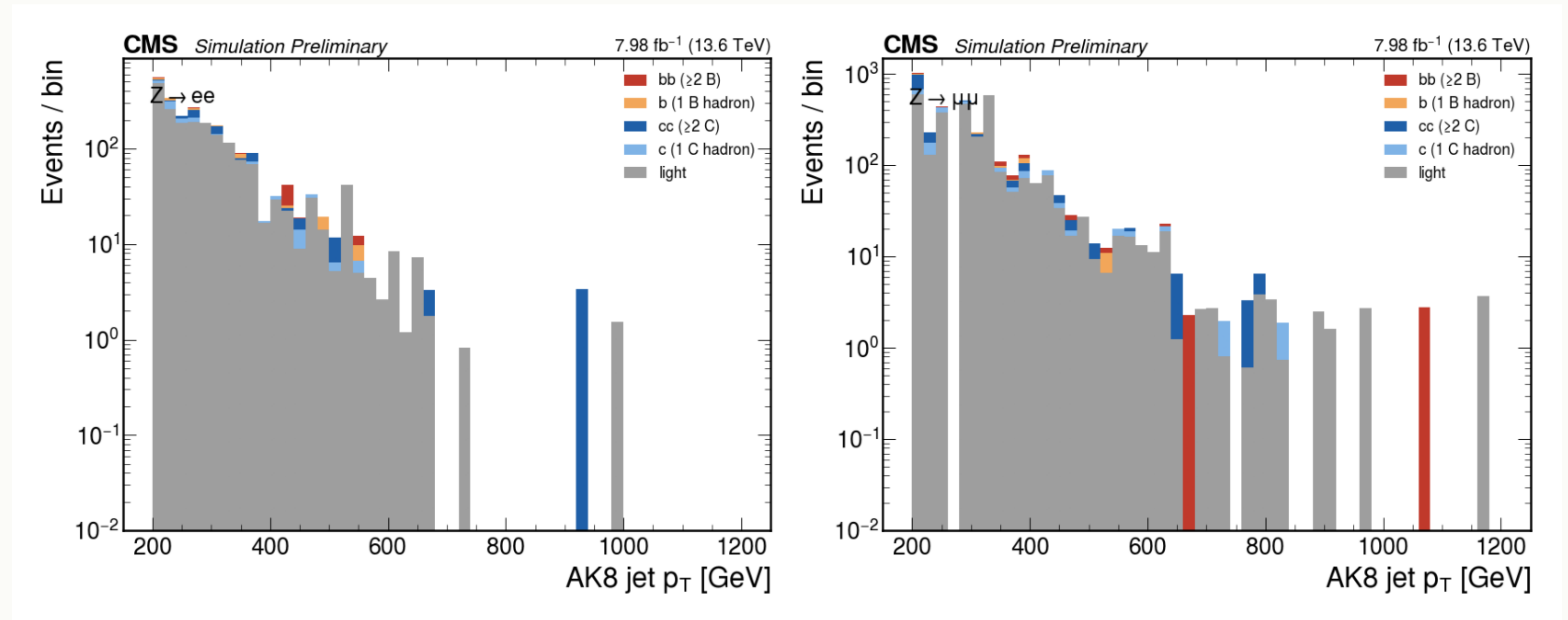
Subleading-lepton p_T

Subleading lepton of the Z candidate, after the full selection ($p_T > 25$ GeV required).



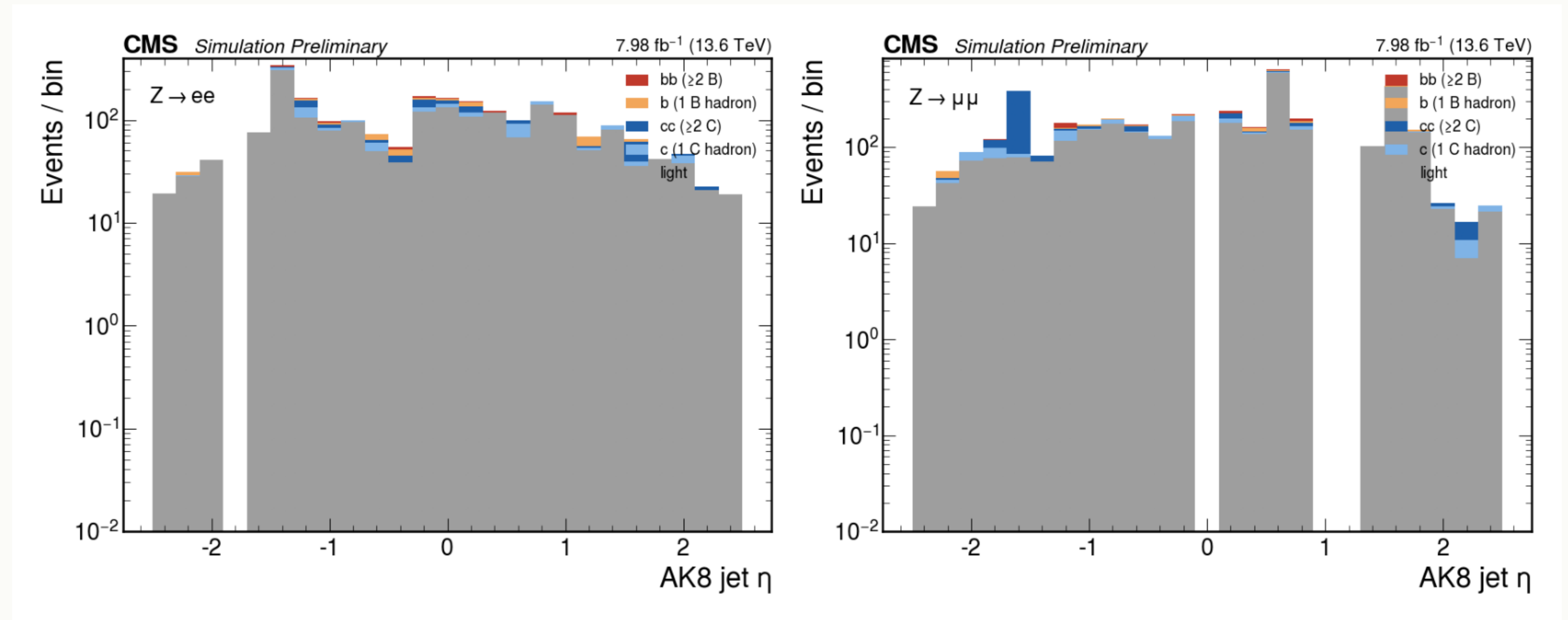
Leading AK8 jet pT

Leading lepton-cleaned AK8 jet with tight ID and two subjects (pT > 200 GeV required).



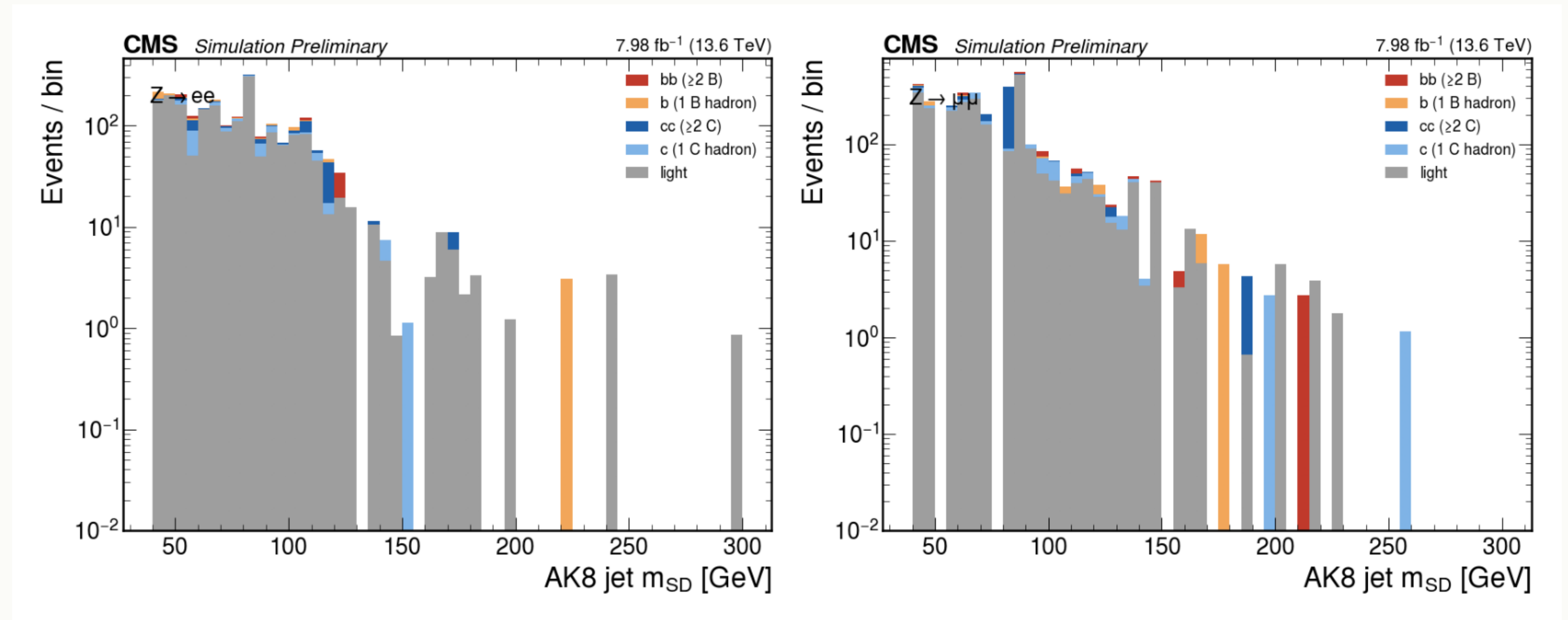
Leading AK8 jet η

Pseudorapidity of the selected AK8 jet ($|\eta| < 2.5$ required).



Leading AK8 jet soft-drop mass

mSD of the selected AK8 jet (mSD > 40 GeV required).



AK8 jet multiplicity

Number of lepton-cleaned AK8 jets with tight ID and two subjects, in events passing the full selection.

